

Yuzheng Hu

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EDUCATION	University of Illinois at Urbana-Champaign <i>Ph.D. in Computer Science</i> ◦ Advisor: Han Zhao Peking University <i>B.S. in Mathematics (Math and Applied Math, Honor Track)</i> ◦ Advisors: Wei Lin and Liwei Wang	Urbana, IL, USA Aug. 2021 – May. 2026 Beijing, China Sep. 2017 – Jul. 2021
RESEARCH INTERESTS	I develop the science of data —a perspective that views data as an <i>optimizable</i> , <i>expandable</i> and <i>predictable</i> component of AI systems. To this end, I work on data attribution , synthetic data , and privacy .	
PUBLICATIONS	★ stands for equal contribution or alphabetical order	
	[P1] ACTG-ARL: Differentially Private Conditional Text Generation with RL-Boosted Control 📄 🔗 Yuzheng Hu , Ryan McKenna, Da Yu, Shanshan Wu, Han Zhao, Zheng Xu, Peter Kairouz <i>The 43rd International Conference on Machine Learning (ICML 2026)</i>	
	[P2] OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pre-training in Every Iteration 📄 😊 Shaobo Wang, Xuan Ouyang, Tianyi Xu, Yuzheng Hu , Jialin Liu, Guo Chen, Tianyu Zhang, Junhao Zheng, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang <i>The 43rd International Conference on Machine Learning (ICML 2026, Spotlight)</i>	
	[P3] A benchmark of expert-level academic questions to assess AI capabilities 📄 🌐 📚 Dataset contributor of Humanity’s Last Exam; authored 4 of 2,500 expert-level questions, with one selected as a prize-winning entry (top 550 worldwide) <i>Nature (2026)</i>	
	[P4] A Snapshot of Influence: A Local Data Attribution Framework for Online Reinforcement Learning 📄 🔗 Yuzheng Hu* , Fan Wu*, Haotian Ye, David Forsyth, James Zou, Nan Jiang, Jiaqi W. Ma, Han Zhao <i>The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Oral)</i>	
	[P5] Taming Hyperparameter Sensitivity in Data Attribution: Practical Selection Without Costly Retraining 📄 Weiyi Wang, Junwei Deng, Yuzheng Hu , Shiyuan Zhang, Xirui Jiang, Runting Zhang, Han Zhao, Jiaqi W. Ma <i>The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)</i>	
	[P6] MergeBench: A Benchmark for Merging Domain-Specialized LLMs 📄 🔗 Yifei He, Siqi Zeng, Yuzheng Hu , Rui Yang, Tong Zhang, Han Zhao <i>The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Datasets and Benchmarks Track)</i>	
	[P7] Empirical Privacy Variance 📄 🔗 Yuzheng Hu* , Fan Wu*, Ruicheng Xian, Yuhang Liu, Lydia Zakyntinou, Pritish Kamath, Chiyuan Zhang, David Forsyth <i>The 42nd International Conference on Machine Learning (ICML 2025)</i>	
	[P8] An Improved Autoregressive Evaluation Paradigm for Large Language Models 📄 Jipeng Zhang*, Rui Pan*, Yuzheng Hu* , Kashun Shum*, Guanyu Yao, Xiang Liu, Renjie Pi, Hanze Dong, Shizhe Diao, Yong Lin, Han Zhao, Tong Zhang <i>ACM Transactions on Intelligent Systems and Technology (ACM TIST 2025)</i>	

- [P9] Most Influential Subset Selection: Challenges, Promises, and Beyond  
Yuzheng Hu, Pingbang Hu, Han Zhao, Jiaqi W. Ma
The 38th Annual Conference on Neural Information Processing Systems (NeurIPS 2024)
- [P10] Localize-and-Stitch: Efficient Model Merging via Sparse Task Arithmetic  
Yifei He, **Yuzheng Hu**, Yong Lin, Tong Zhang, Han Zhao
Transactions on Machine Learning Research (TMLR 2024)
- [P11] SoK: Privacy-Preserving Data Synthesis  
Yuzheng Hu*, Fan Wu*, Qinbin Li, Yunhui Long, Gonzalo Munilla Garrido, Chang Ge, Bolin Ding, David Forsyth, Bo Li, Dawn Song
The 45th IEEE Symposium on Security and Privacy (S&P 2024)
- [P12] HPL-ViT: A Unified Perception Framework for Heterogeneous Parallel LiDARs in V2V 
Yuhang Liu, Boyi Sun, Yuke Li, **Yuzheng Hu**, Fei-Yue Wang
The 41st IEEE International Conference on Robotics and Automation (ICRA 2024)
- [P13] Revisiting Scalarization in Multi-Task Learning: A Theoretical Perspective 
Yuzheng Hu, Ruicheng Xian, Qilong Wu, Qiuling Fan, Lang Yin, Han Zhao
The 37th Annual Conference on Neural Information Processing Systems (NeurIPS 2023)
- [P14] Understanding the Impact of Adversarial Robustness on Accuracy Disparity  
Yuzheng Hu, Fan Wu, Hongyang Zhang, Han Zhao
The 40th International Conference on Machine Learning (ICML 2023)
- [P15] Actor-critic is implicitly biased towards high entropy optimal policies 
Yuzheng Hu*, Ziwei Ji*, Matus Telgarsky*
The 10th International Conference on Learning Representations (ICLR 2022)
- [P16] Towards Understanding the Data Dependency of Mixup-style Training  
Muthu Chidambaram, Xiang Wang, **Yuzheng Hu**, Chenwei Wu, Rong Ge
*The 10th International Conference on Learning Representations (ICLR 2022, **Spotlight**)*
- [P17] Don't Waste Your Bits! Squeeze Activations and Gradients for Deep Neural Networks via TinyScript 
Fangcheng Fu, **Yuzheng Hu**, Yihan He, Jiawei Jiang, Yingxia Shao, Ce Zhang, Bin Cui
The 37th International Conference on Machine Learning (ICML 2020)

- MANUSCRIPTS [M1] MAPLE: Metadata Augmented Private Language Evolution 
Eli Chien, **Yuzheng Hu**, Ryan McKenna, Shanshan Wu, Zheng Xu, Peter Kairouz
- [M2] A Unified Theory of Random Projection for Influence Functions  
Pingbang Hu*, **Yuzheng Hu***, Jiaqi W. Ma*, Han Zhao*
- [M3] A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI 
Junwei Deng*, **Yuzheng Hu***, Pingbang Hu*, Ting-Wei Li*, Shixuan Liu*, Jiachen T. Wang, 21 others, Jiaqi W. Ma

INVITED TALKS **Differentially Private Conditional Text Generation with RL-Boosted Control**

- Google Athena seminar Sep. 2025
- Google ML Privacy seminar (Host: Natalia Ponomareva) Sep. 2025

Empirical Privacy Variance

- Google ML Privacy seminar (Host: [Matthew Jagielski](#)) Mar. 2025

INTERNSHIP EXPERIENCE	Google Research	Mountain View, CA, US
	<i>Research Intern, Hosts: Zheng Xu, Ryan McKenna, Da Yu; Manager: Peter Kairouz</i>	May. 2025 – Nov. 2025
	◦ Developed a metadata-conditioned framework for DP synthetic text generation with [P1] and without [M1] fine-tuning	
	Jane Street Group, LLC	New York, NY, US
	<i>Quantitative Trading Intern</i>	May. 2024 – July. 2024
	◦ Ranked 2/200 (1/150 among interns) in the Tenth Annual Figgie World Championships ◦ Participated in mock trading games and worked on two desk projects (options and equities)	
	Damo Academy, Alibaba Group (U.S.) Inc.	Bellevue, WA, US
	<i>Research Intern, Mentor: Bolin Ding</i>	May. 2022 – Aug. 2022
	◦ Worked on the intersection of privacy and law ◦ Wrote a systematization of knowledge paper on privacy-preserving data synthesis [P11]	
RESEARCH EXPERIENCE	Simons Institute for the Theory of Computing	Berkeley, CA, US
	<i>Visiting Graduate Student, Program: Modern Paradigms in Generalization</i> 	Aug. 2024 – Dec. 2024
	◦ Quantified the discrepancy between differential privacy and memorization in LLMs [P7], with Pritish Kamath and Chiyuan Zhang	
	University of Illinois at Urbana-Champaign	Urbana, IL, US
	<i>Graduate Research Assistant, Advisor: Han Zhao</i>	Feb. 2022 – now
	◦ Thesis research: data attribution [P4] [P5] [P9] [M2] [M3], with Jiaqi Ma ◦ Other works: model merging [P6] [P10]; representational limitations of scalarization in multi-task learning [P13]; trade-off between robustness and fairness [P14]	
	University of Illinois at Urbana-Champaign	Urbana, IL, US
	<i>Graduate Research Assistant, Advisor: Matus Telgarsky</i>	Aug. 2021 – Dec. 2021
	◦ Proved that actor-critic converges to high entropy optimal policy without explicit regularization [P15]	
	Duke University	Remote
	<i>Undergraduate Research Assistant, Advisor: Rong Ge</i>	Jun. 2020 – Feb. 2021
	◦ Demonstrated the data-dependency of mixup [P16]	
	School of Intelligence Science and Technology, Peking University	Beijing, China
	<i>Undergraduate Research Assistant, Advisor: Liwei Wang</i>	Aug. 2019 – Jun. 2020
	◦ Identified two non-local properties of the loss landscape of neural networks	
	School of Computer Science, Peking University	Beijing, China
	<i>Undergraduate Research Assistant, Advisor: Bin Cui</i>	Aug. 2018 – Aug. 2019
	◦ Introduced a distribution-aware, non-uniform quantization algorithm for multilayer perceptrons [P17]	
TEACHING EXPERIENCE	UIUC CS 591 – Machine Learning Reading Group	Spring 2024, Fall 2024
	UIUC CS 598 – Transfer Learning	Spring 2023
	UIUC CS 498 – Trustworthy Machine Learning	Fall 2022
SERVICES	Seminar Co-Organizer: UIUC Machine Learning Seminar	2024 – 2026
	Workshop Organizer and Primary Contact: DATA-FM @ ICLR 2026	
	Area Chair: NeurIPS 2026 Position Paper Track	
	Program Committee: TPD (2026, 2025), ICLR SynthData (2025) , ACL L2M2 (2025)	
	Reviewer: NeurIPS (2022-2025), ICML (2023-2026), ICLR (2024-2026), AISTATS (2024), TMLR	

HONORS AND AWARDS	Top/Outstanding Reviewer: ICML 2025 (Top 2%), NeurIPS 2025 (Top 8%), NeurIPS 2023 (Top 10%)	
	NeurIPS 2025 Oral (Top 0.3%, 77/21,575)	2025
	Saburo Muroga Endowed Fellowship (Top 4% in UIUC, 4/100+)	2021-2022
	Exceptional Award for Academic Innovation (Top 1% in Peking University)	2020
	Elite Undergraduate Training Program of Applied Math (Top 15% in SMS)	2019-2021
	Invited Participant, China TST for the 58th International Mathematical Olympiad (IMO)	2017
	Gold Medal, 32nd Chinese Mathematical Olympiad (CMO) (Top 60 in China)	2016
TRAVEL GRANTS	TPDP Student Travel Funding (\$825)	2025
	IEEE S&P Student Travel Grants Award (\$1000)	2024
	NeurIPS 2023 Scholar Award (\$2000)	2023
SKILLS	Programming	Python, Shell Script, $\LaTeX$, MATLAB
	Frameworks & Packages	PyTorch, HuggingFace